

**1. Which of the following has little effect on pancreatic secretion or its components?**

- A. Acetylcholine.
- B. Sympathetic stimulation**
- C. Secretin
- D. Cholecystokinin
- E. VIP

**2-. Defective parietal cells would result in malabsorption of which vitamin?**

- A. Vitamin B1
- B. Vitamin B2
- C. Niacin
- D. Vitamin B12**
- E. Folic acid.

**3- Of the approximately 9 liters of water that go through the GI tract daily, how much makes it out in the feces?**

- A. 100 ml**
- B. 1000 ml
- C. 2000 ml
- D. 1500 ml
- E. 500 ml

**4. The progressive wave of muscle contractions that proceeds along the esophagus, compressing the lumen and forcing food ahead of it is called:**

- A. primary peristalsis**
- B. segmentation
- C. the migrating myoelectric complex
- D. mass movement

**5. Vitamin B12 is absorbed mainly in the:**

- A. stomach
- B. ileum**
- C. colon
- D. jejunum

**6. Saliva contains the carbohydrate-digesting enzyme:**

- A. pepsin
- B. carboxypeptidase
- C. amylase**
- D. trypsin
- E. lipase

**7. Which is the primary absorptive organ of the digestive system?**

- A) small intestine**
- B) stomach
- C) colon
- D) pancreas

**16. Which tissue layer provides for primary digestive motility?**

- A) Mucosa
- B) submucosa
- C) mesentery
- D) muscularis mucosa

**18. Proteins are primarily digested to and absorbed as...**

- A) amino acids
- B) monosaccharides
- C) nucleic acids
- D) fatty acids & glycerol

**19. HCL secretion by the stomach...**

- A) Is triggered by chewing
- B) denatures proteins
- C) activates enzymes that digest proteins
- D) all of these

**20. What prevents food from entering the nasal passages during swallowing?**

- A) elevation of uvula
- B) contraction of pharyngeal muscles
- C) apposition of vocal folds
- D) elevation of epiglottis

**23. The major chemical digestive activity that takes place in the stomach is...**

- A) breakdown of starch
- B) breakdown of proteins
- C) digestion of fats
- D) neutralization of acid by buffers & mucous

**24. Bile acts on...**

- A) distention in the stomach
- B) acid in the stomach
- C) fat in the stomach
- D) fat in the duodenum

**25. Chief cells of gastric mucosa secrete...**

- A) Trypsin
- B) bicarbonate ions
- C) pepsinogen
- D) HCL

**26. Which of the following is true about pancreatic secretions?**

- A) CCK causes release of bicarbonate
- B) secretin causes release of enzymes
- C) gastrin stimulates release of enzymes
- D) CCK stimulates release of enzymes while secretin stimulates release of bicarbonate

**27. The hormone that inhibit gastric secretion and motility is...**

- A) Gastrin
- B) histamine
- C) secretin
- D) Pepsin

**28. Intrinsic factor is...**

- A) secreted by parietal cells and is responsible for vitamin B12 absorption
- B) not necessary for normal body function
- C) abundant in pernicious anemia
- D) an aminopeptidase

**29. The ileocecal sphincter:**

- A) Prevents acid from entering the esophagus
- B) prevents bacterial growth in the stomach
- C) prevents bacterial infection of small intestine
- D) inhibits gastric emptying

**30. The intrinsic nerve plexuses...**

- A) are located in the mucosal region
- B) network of neurons that act as "gut" brain
- C) are made up of two plexuses
- D) B+C

**31. Lipids:**

- A) are broken down by pepsin
- B) are absorbed into lacteals
- C) become part of micelles in the duodenum
- D) B+C

**33. Which of the following is the main digestible carbohydrate normally consumed in the human diet?**

- A) Starch
- B) Cellulose
- C) Maltose
- d) amino acid

**35. Stimulation of the submucosal plexus results in an increase in which of the following?**

- A) Stomach PH
- B) Secretion of the gut
- C) Motility of the gut
- d) non of above

**36. Inhibition of vagal function has the greatest effect on which segment of the alimentary tract?**

**A) Anal sphincter**

- B) Descending colon
- C) stomach**
- d) rectum

**38. Which of the following is true of the gastro enteric reflex?**

- a) It leads to defecation
- b) It stimulates contractions of the colon
- c) It leads to increased peristaltic activity in the small intestine**
- d) non of above

**39. Which of the following decreases blood flow to the villi of the small intestine?**

- a) adrenalin
- B) Norepinephrine**
- c) acetylcholine
- d) HCL

**41. Sodium export through the brush border of the intestinal epithelial cells occurs by which of the following mechanisms?**

- A): passive diffusion
- B) Facilitated diffusion**
- c) simple diffusion
- d) non of above

**42. Which of following digestive organs does not show peristalsis:**

- A) Esophagus
- B) Stomach
- C) Small intestined
- D) Caecum**

**43. Regarding slow-waves:**

- A) Cause Muscle contractions by themselves.
- B) Are Action potential
- C) Cause Sodium Ions to Enter.
- D) Cause by interaction between smooth muscle cells and interstitial cells of Cajal.
- E) Both C+D**

**44-The physiological actions of CCK include:**

- a) Increased oesophageal motility
- b) Closing the sphincter of oddi
- c) Increased bile synthesis
- d) Contraction of pancreatic acini
- e) Increased enzyme secretion from the pancreas**

**45. The component parts of the myenteric plexus of the enteric nervous system include the following except:**

- a) Sensory neurons
- b) Interneurons

- c) Motor neurons
- d) Synapses
- e) Control centers in the medulla oblongata

**46. Saliva does not contain:**

- a) Blood group antigens
- b) Lysozymes
- c) Immunoglobulins
- d) Platelets
- e) Chloride

**47. The functions of saliva include the following except:**

- a) Cleaning the mouth
- b) Killing bacteria
- c) Digestion of proteins
- d) Facilitation of speech
- e) Digestion of lipids

**48. The pharyngeal phase of swallowing:**

- a) Is partially voluntary
- b) Includes opening of the upper oesophageal sphincter
- c) Includes the peristaltic activity of the upper third of the oesophagus
- d) Is associated with the opening of the glottis
- e) Is preceded by deep inspiration

**49. The enzymes produced by the exocrine pancreas include the following except:**

- a) Proelastase
- b) DNase
- c) Amylase
- d) Enteropeptidase
- e) Lipase

**50. The secretion of the exocrine pancreas is increased by the following except:**

- a) Vagal stimulation
- b) Acetylcholine
- c) Secretin
- d) CCK
- e) Atropine

**51. Bile production by the liver cells is increased by:**

- a) Sympathetic activation
- b) Secretin
- c) CCK
- d) Bile salts
- e) Atropine

**52. The jejunum is the main site for absorption of the following except:**

- a) Glucose

- b) Amino acids
- c) Fatty acids
- d) Bile salts
- e) Water

**53. The functions of the colon include the following except:**

- a) Absorption of water
- b) Absorption of sodium
- c) Absorption of chlorid
- d) Absorption of calcium
- e) Absorption of potassium

**54. The intestinal flora is:**

- a) Mainly viruses
- b) Mainly found in the duodenum
- c) Needed for normal digestion
- d) Mainly found in the colon
- e) Mainly pathogenic bacteria

**55. The following is true about the small intestine:**

- a) Stores food for several hours
- b) Is the main source of digestive enzymes
- c) Has haustrae
- d) contractions Has segmentation
- e) Secretes the intrinsic factors

**56. The defaecation reflex:**

- a) Is initiated by stretching the anal canal
- b) Is integrated in a control center in the lumbar spinal cord control center in the Is integrated in a c
- c) sacral spinal cord
- d) Is controlled by the vagus nerve
- e) Involves contraction of the sigmoid colon

**57. Which of the following is likely to fail to reduce secretion of hydrochloric acid in the stomach:**

- a) A H<sub>2</sub> receptor blocker
- b) A nicotinic cholinergic receptor blocker
- c) A muscarinic cholinergic receptor blocker
- d) An inhibitor of the hydrogen-potassium pump
- e) A gastrin receptor blocker

**58. The following statements are true about bile except:**

- a) 25% is produced by bile duct cells
- b) The bile stored in the gall bladder contains secondary bile acids
- c) Its secretion is increased by secretin
- d) Bile pigments are needed for micell formation
- e) Some lipids will be digested and absorbed in the absence of bile

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- a) Sensory neurons
- b) Interneurons
- c) Motor neurons
- d) Synapses
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**60. The secretion of the exocrine pancreas is increased by the following except:**

- a) Vagal stimulation
- b) Acetylcholine
- c) Secretin
- d) CCK
- e) Noradrenaline

**61. The jejunum is the main site for absorption of except:**

- a) Glucose
- b) Amino acids
- c) Fatty acids
- d) Vitamin B12
- e) Water

**62. Functions of the human stomach include:**

- a) Absorption of iron
- b) Absorption of Vitamin B12
- c) Absorption of 50% of water intake
- d) Storage of food for 24 hours
- e) Limited digestion of proteins

**63. Relating to carbohydrate digestion:**

- a) Cellulose can be digested by human amylases
- b) About one third of starch digestion occurs in the mouth and two-thirds occurs in the small gut
- c) It can only occur in the complete absence of protein digestion
- d) It occurs only if bile salts are present
- e) Sucrose is split by amylase action

**64. Contraction of the gallbladder may be due to:**

- a) Secretin
- b) VIP
- c) Bile salts
- d) Atropine
- e) Parasympathetic stimulation

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**65. Saliva does not contain:**

- a) Blood group antigens
- b) Lysozymes
- c) Immunoglobulins
- d) Pepsin
- e) Chloride

**66. The actions of secretin include:**

- a) Increased salivary secretion
- b) Relaxation of the oesophageal sphincters
- c) Stimulation of gastric secretion
- d) Stimulation of secretion of bicarbonate by pancreatic duct cells
- e) Stimulation of secretion of bicarbonate by pancreatic acinar cells

**67. Saliva is needed for:**

- a) Digestion of sucrose
- b) Digestion of phospholipids
- c) Ability to speak
- d) Breaking food down into small pieces
- e) Absorption of chloride

**68-Major constituents of feces include all of the following except:**

- a) Water
- b) Cellulose
- c) Bacteria
- d) Calcium salts
- e) Glycogen

**69. The exocrine pancreas secretes the following enzymes except:**

- a) Chymotrypsinogen
- b) Proleptase
- c) Lactase
- d) Alpha-amylase
- e) Phospholipase

**70-GIT hormones with a major effect on the stomach include the following except:**

- a) Gastrin
- b) Secretin
- c) GIP
- d) Neurotensin
- e) CCK